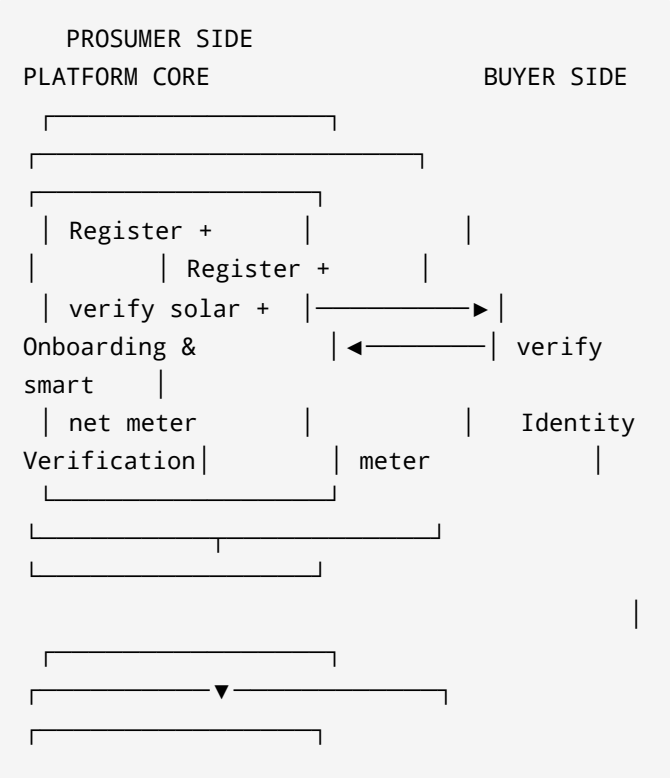
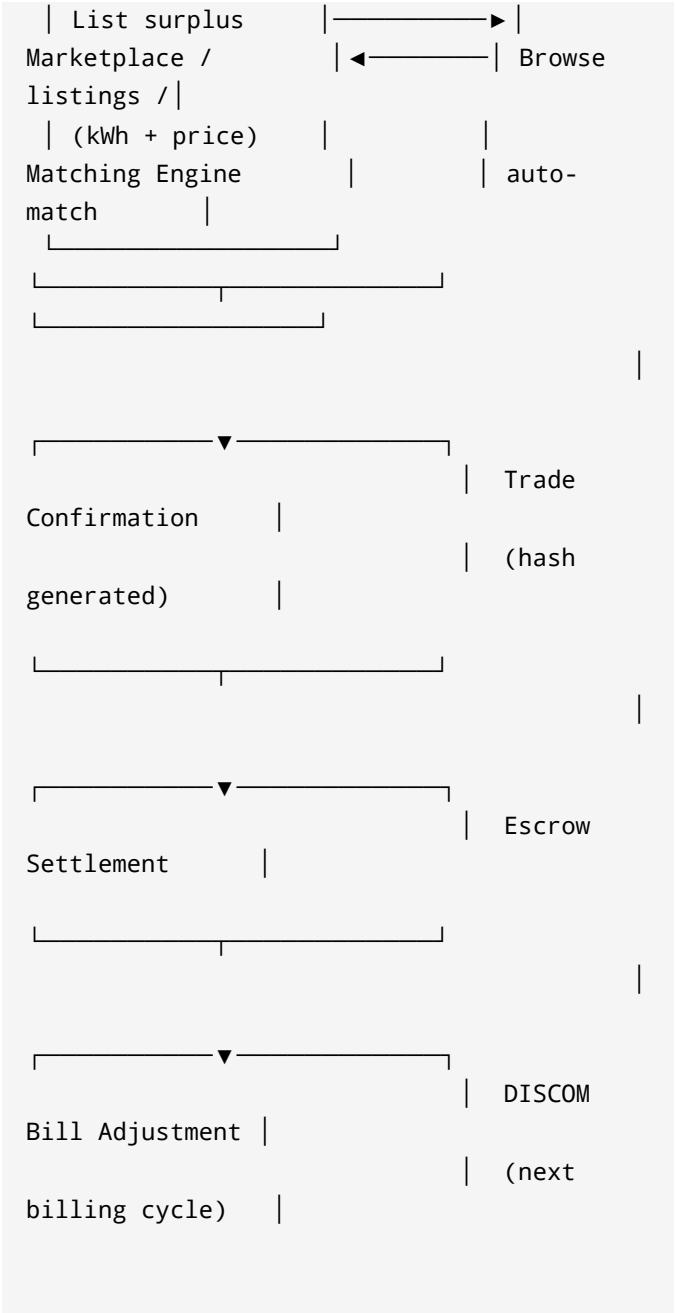
SolarShare — App Flow & User Journey

Version: 1.0 | **Companion to:** PRD & TRD v1.0

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1. High-Level App Flow




List surplus
Marketplace / listings
(kWh + price)
Browse
Matching Engine
auto-match
Trade Confirmation
(hash generated)
Escrow Settlement
DISCOM Bill Adjustment
(next billing cycle)

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2. Detailed User Journey — Prosumer ("Priya")

Step	Action	System Response
1	Downloads app / opens WhatsApp bot	Requests DISCOM account number + net meter ID for verification
2	Verification completes	Dashboard shows today's live generation vs. consumption
3	Surplus detected (e.g., 4.2 kWh)	App suggests a listing price (shown against feed-in tariff and retail tariff for context)
4	Confirms listing	Listing becomes visible to nearby verified buyers
5	Match found	Push/WhatsApp notification: "2 kWh

Step	Action	System Response
		matched with Buyer #114 at ₹6/unit"
6	Trade auto-confirms (or manual approval, configurable)	Transaction hash generated; added to ledger
7	End of billing cycle	Sees earnings summary + CO_2 impact in the app

3. Detailed User Journey — Buyer ("Rahul")

Step	Action	System Response
1	Downloads app / opens WhatsApp bot	Requests DISCOM account number + smart meter ID
2	Verification completes	Dashboard shows nearby available surplus (if any)
3	Browses listings or enables auto-buy	Sees price comparison: P2P

Step	Action	System Response
		rate vs. standard retail tariff
4	Confirms purchase	Trade hash generated; escrow holds payment
5	Energy "delivered" (accounting, not physical routing — grid handles actual flow)	Confirmation notification
6	No local surplus available at a given time	App transparently states buyer is served by standard grid — no service interruption
7	End of billing cycle	Sees savings summary vs. standard retail billing

4. Detailed User Journey — Society Admin ("RWA Coordinator")

Step	Action	System Response
1	Onboards as verified society administrator	Gains access to a society-level dashboard
2	Invites residents (bulk WhatsApp invite)	Tracks onboarding completion rate
3	Views society-wide trading activity	Sees total kWh traded, CO_2 avoided, top contributors (leaderboard)
4	Handles disputes/escalations (rare)	Access to trade history with hash IDs for verification

5. Key Edge Cases

Scenario	Expected Behavior
No buyer matches a seller's surplus	Surplus defaults to standard net-metering export; seller still gets DISCOM feed-in rate, no value lost

Scenario	Expected Behavior
No seller surplus available for a buyer	Buyer seamlessly draws from grid at standard retail rate; app clearly communicates "grid-supplied" status
Cloudy/low-generation day	Dashboard shows reduced/zero surplus prediction in advance so sellers aren't caught off guard
Meter verification fails	User blocked from listing/buying until DISCOM re-verification; prevents fraudulent listings
Trade dispute (rare, since settlement is escrow-based)	Both parties can view the immutable, timestamped hash record as evidence

6. Core Design Principle Behind the Flow

Every step is designed so that **the grid never stops working** — SolarShare only ever adds a better-priced option on top of the existing, always-

available DISCOM supply. This is why the "fallback to grid" path appears explicitly at multiple points in the flow rather than being treated as a failure state.